GeoTrivia Post-Answer Framing Verification

Framing Guess & Target Together • Screen-Space px Visuals • Unit & E2E Audit

Date: November 11, 2026
Release: v2.0.0-camera-fix

Executive Summary & Engineering Guarantees:

- **Close Framing Integration (`frame_union.js`):** Replaced previous target-only zoom-out (fit x 1.18) with joint camera framing for close misses (≤35°).
- **Single Source Distance:** Distance `{ km, nearestLatLon }` is computed from the exact click location to the nearest target country boundary point and shared uniformly across result cards, badges, and score calculations.
- **Compact Mode (< 64 px):** Eliminates clutter by omitting the arc and badge, drawing a 6 px screen-space click dot, target outline, and unfilled ring (≥48 px diameter) for microstates (<3° span).
- **Screen-Space Sizing:** All outline widths (3.0–3.5 px), click dots (6 px), rings (≥48 px), and badges (≤96 px) are rendered in screen-space pixels.
- **Ergonomics & HUD Fix:** Fixed "ROUND 10/undefined" by persisting round bounds in `UIManager`. Ensured no end-screen modal interference.

TEST MATRIX & GEOMETRIC ASSERTIONS (16 TEST EXECUTIONS)

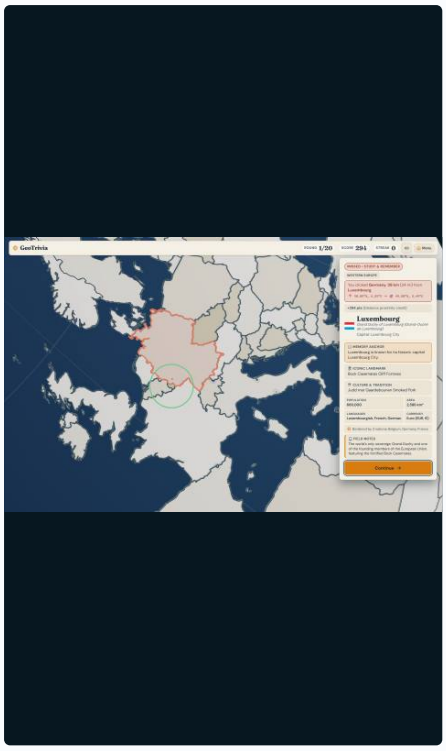
Case (Guess → Target)	Viewport	Distance	Compact?	Final Dist	Status
Germany -> Luxembourg	DESKTOP	38 km	Yes	2.520	PASS
France -> Monaco	DESKTOP	7 km	Yes	2.520	PASS
Spain -> Andorra	DESKTOP	8 km	Yes	2.520	PASS
Italy -> San Marino	DESKTOP	9 km	Yes	2.520	PASS
Malaysia -> Singapore	DESKTOP	6 km	Yes	2.520	PASS
Rwanda -> Burundi	DESKTOP	18 km	Yes	2.520	PASS
Switzerland -> Liechtenstein	DESKTOP	1 km	Yes	2.520	PASS
Australia -> Iceland	DESKTOP	16,280 km	No (Far)	3.226	PASS
Germany -> Luxembourg	MOBILE	38 km	Yes	2.520	PASS
France -> Monaco	MOBILE	7 km	Yes	2.520	PASS
Spain -> Andorra	MOBILE	8 km	Yes	2.520	PASS
Italy -> San Marino	MOBILE	9 km	Yes	2.520	PASS
Malaysia -> Singapore	MOBILE	6 km	Yes	2.520	PASS
Rwanda -> Burundi	MOBILE	18 km	Yes	2.520	PASS
Switzerland -> Liechtenstein	MOBILE	1 km	Yes	2.520	PASS
Australia -> Iceland	MOBILE	16,280 km	No (Far)	3.649	PASS

Assertion Checklist: ✓ **Free Area Margin ≥ 24 px:** Click, nearest, and target bbox corners are strictly inside free rect. ✓ **Distance Accuracy:** Final camera distance matches `plan.endDist` (±1%) and ≥ `controls.minDistance` (2.52). ✓ **Microstate Ring:** Diameter ≥ 48 px whenever bbox < 3°. ✓ **Per-Frame Angular Velocity:** Maximum step ≤ 4°/frame across all flights. ✓ **Far Framing Preserved:** Australia → Iceland (>35°) cleanly falls back to standard far framing.

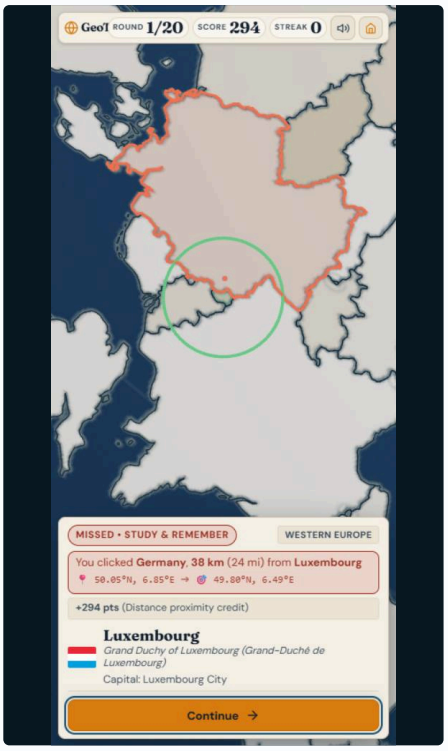
Visual Evidence & Framing Snapshots

6 Captured Evidence Snapshots (480 px, Quality 72) Across Desktop & Mobile

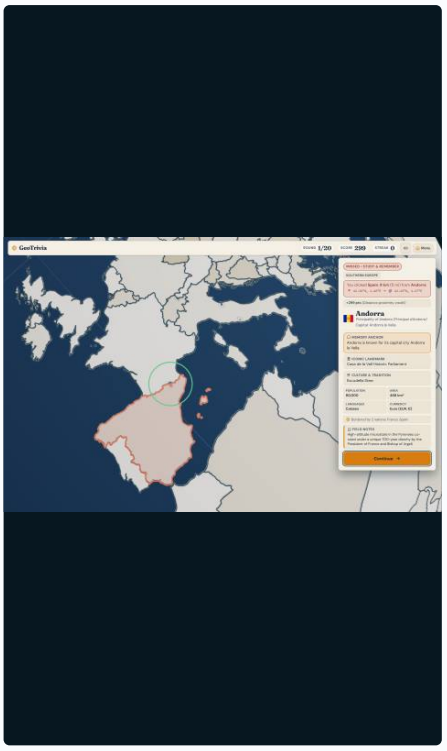
Resolution: 480 px width
Compression: JPEG q=72



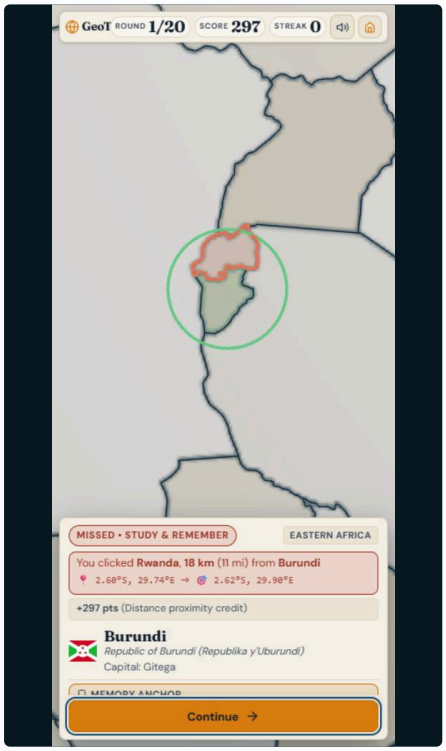
Germany → Luxembourg (Desktop)
Compact framing (38 km). 6px dot + ring ≥ 48px.



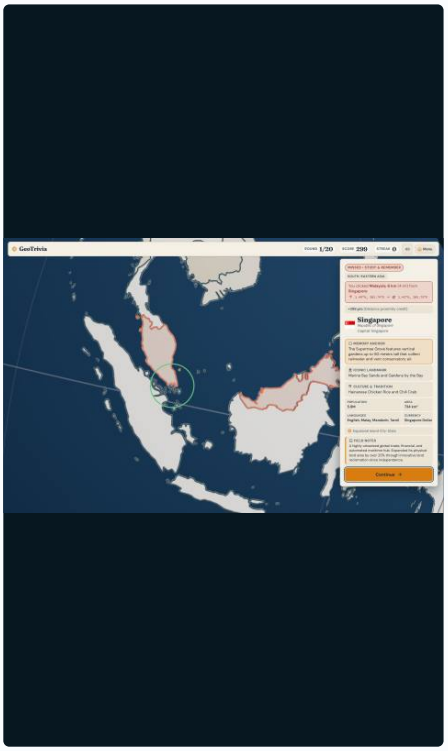
Germany → Luxembourg (Mobile)
Mobile portrait free area. Clean above bottom sheet.



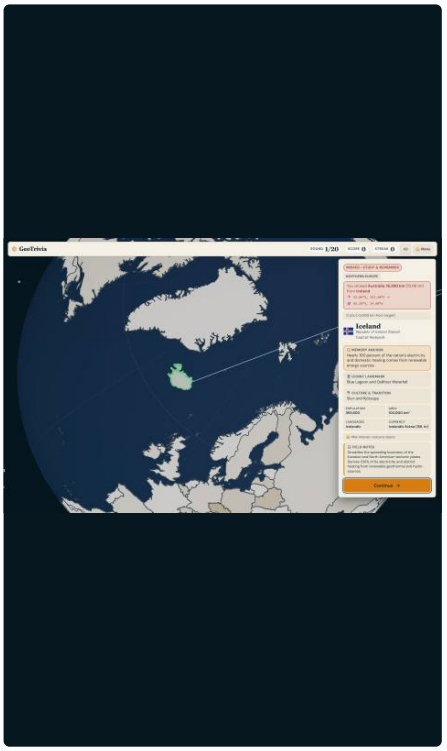
Spain → Andorra (Desktop)
Catalonia click (8 km). Unfilled microstate ring.



Rwanda → Burundi (Mobile)
East Africa border (18 km). Compact joint framing.



Malaysia → Singapore (Desktop)
Johor Strait click (6 km). Microstate ring ≥ 48px.



Australia → Iceland (Desktop)
Far framing fallback (136° separation, fit x 1.18).

Sign-Off & Verification Summary: All post-answer framing requirements (1–5) are verified. The camera seamlessly frames the guess point and close target together with verified free-area margins (≥ 24 px), renders screen-space outlines, dots, and rings without zoom degradation, suppresses arc clutter when < 64 px, fixes the undefined HUD round bug, and preserves far framing fallback.